



Mark Scheme (Results)

January 2023

Pearson Edexcel International Advanced
Subsidiary Level in Chemistry (WCH13)
Paper 01 Practical Skills in Chemistry I

Question Number	Answer	Additional Guidance	Mark
1(a)	An answer that makes reference to the following points: - Alkene - hex-1-ene, (<i>E/Z</i>,cis/trans)hex-2-ene, (<i>E/Z</i>,cis/trans)hex-3-ene	(1) Allow C=C / carbon-carbon double bond Ignore ene Ignore just “double bond” (1) Allow hexene /1-hexene etc	(2)

Question Number	Answer	Additional Guidance	Mark
1(b)	An answer that makes reference to the following points: - K⁺ / potassium - I⁻ / iodide	(1) Do not award K (1) Do not award just I / iodine Allow one mark for just KI Ignore state symbols Ignore “iodine ion”	(2)

Question Number	Answer	Additional Guidance	Mark
1(c)(i)	An answer that makes reference to the following points: - balanced ionic equation - state symbols	(1) $2\text{I}^-(\text{aq}) + \text{Br}_2(\text{aq}) \rightarrow \text{I}_2(\text{aq}) + 2\text{Br}^-(\text{aq})$ Allow multiples Do not award uncanceled ions on both sides (1) Allow one mark for correct state symbols for an unbalanced / incorrect species equation	(2)

Question Number	Answer	Additional Guidance	Mark
1(c)(ii)	An answer that makes reference to the following points: - two layers would form - the organic / colourless layer would change to purple / violet (and then decolourises slowly) - the aqueous / red-brown layer would become colourless / fade / yellow	(1) This mark can be awarded from a description of colour changes of two layers. (1) Accept top layer would change to purple / violet Do not award pink (1) Accept bottom layer would become colourless / fade / yellow If no layers stated then allow one mark for decolorised In M2 and M3 Do not award precipitate formed/goes cloudy Do not award any gas/fumes given off /bubbles Penalise once only	(3)

(Total for Question 1 = 9 marks)

Question Number	Answer	Additional Guidance	Mark
2(a)(i)	An answer that makes reference to the following point: reducing the heat loss	Allow reference to ensuring the flame is stable Allow ensuring more of the energy from the flame is transferred to the water Ignore just reducing/stopping draughts Ignore prevent heat loss	(1)

Question Number	Answer	Additional Guidance	Mark
2(a)(ii)	An answer that makes reference to the following point: 	Allow any recognisable flame Ignore lack of line Do not award oxidising symbol 	(1)

Question Number	Answer	Additional Guidance	Mark
2(b)(i)	An answer that makes reference to the following points: 7.3(°C) and 0.32(g)		(1)

Question Number	Answer	Additional Guidance	Mark
2(b)(ii)	- calculation of energy transferred to water (1) - calculation of moles ethanol burned (1) - calculation of enthalpy of combustion with negative sign and 2/3 SF (1)	Example of calculation: $Q = 150 \times 4.18 \times 7.3 \div 1000 = 4.5771 \text{ (kJ)}$ Allow 4577.1 (J) $0.32 \div 46 = 0.0069565 / 6.9565 \times 10^{-3} \text{ (mol)}$ $4.5771 / 0.0069565 = -657.96$ $= -660 / -658 \text{ (kJ mol}^{-1}\text{)}$ Allow $-660\,000 / -658\,000 \text{ J mol}^{-1}$ Ignore SF except in final answer TE throughout Correct answer with no working scores 3 Do not award incorrect units in final answer only	(3)

Question Number	Answer	Additional Guidance	Mark
2(b)(iii)	An explanation that makes reference to the following points: - some of the ethanol would be burned without heating the water / the mass of ethanol burned would be greater than expected (1) - so final value for enthalpy change of combustion would be less exothermic / less negative (1)	Ignore references to effect on ΔT Ignore just heat loss to the environment Allow lower / less / smaller M2 depends on M1 or near miss	(2)

Question Number	Answer	Additional Guidance	Mark
2(b)(iv)	calculation of percentage uncertainty	Example of calculation: $(2 \times 0.1 \div 7.3) \times 100 = 2.7397 / 2.740 / 2.74 / 2.7 / 3 \%$ Ignore SF	(1)

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	An answer that makes reference to the following point: increasing the temperature (change) / mass of ethanol burned (will improve / reduce the percentage uncertainty)		(1)

Question Number	Answer	Additional Guidance	Mark
2(c)(ii)	An explanation that makes reference to the following points: - uncertainty values indicate the range (over which the measured value is valid) (1) - accuracy is an indication of the difference between the experimental value and the data book value (1)	Allow heat loss greater because the temperature is higher so the experiment will be less accurate	(2)

(Total for Question 2 = 12 marks)

Question Number	Answer	Additional Guidance	Mark
3(a)(i)	An answer that makes reference to the following points: (pipette) lower / decrease (1) (volumetric flask) no change / no effect (1)	Ignore any explanations even if incorrect	(2)

Question Number	Answer	Additional Guidance	Mark
3(a)(ii)	An answer that makes reference to the following point: to ensure that the solution has the same concentration throughout / is homogeneous	Accept to make sure that it is mixed (thoroughly) / uniform Ignore references to dissolving	(1)

Question Number	Answer	Additional Guidance	Mark
3(a)(iii)	An answer that makes reference to the following points: (sodium hydroxide) solution is an irritant / harmful / damages eyes / skin (1) (if filled above head height) the sodium hydroxide might splash into eyes / into face / onto arm (1)	Do not award corrosive / caustic / toxic Allow reverse argument Do not award references to fumes / inhalation	(2)

Question Number	Answer	Additional Guidance	Mark
3(a)(iv)	An answer that makes reference to the following points: (from) colourless / no colour (1) (to) (pale) pink (1)	Correct colours in reverse order scores 1 Do not award red / purple	(2)

Question Number	Answer	Additional Guidance	Mark
3(a)(v)	completion of titrations 2 and 3 and calculation of mean from four concordant values	Example of calculation: (Titrations 2 and 3 28.85 and 28.7(0)) $(28.75 + 28.85 + 28.70 + 28.90) \div 4 = 28.8(0) \text{ (cm}^3\text{)}$	(1)

Question Number	Answer	Additional Guidance	Mark
3(a)(vi)	An answer that makes reference to the following point: titrations 1 and 2 are concordant	Allow titrations 1 and 2 are within 0.2 cm ³ Allow there are already (two) concordant results Accept consistent for concordant	(1)

Question Number	Answer	Additional Guidance	Mark
3(a)(vii)	An answer that makes reference to the following point: otherwise the titres would be greater than the capacity of a burette/ so the titres would not be greater than the volume of the burette	Allow the titre would be too large for the burette / ten times bigger / very large / too much alkali required for an accurate titration	(1)

Question Number	Answer	Additional Guidance	Mark
3(b)(i)	An answer that makes reference to the following points: 3 and 3		(1)

Question Number	Answer	Additional Guidance	Mark
3(b)(ii)	- calculation of moles sodium hydroxide in titre (1) - calculation of moles citric acid in 25 cm³ aliquot (1) - calculation of concentration of citric acid in Solution B (1) - calculation of concentration citric acid in Solution A (1) - calculation of concentration of citric acid in g dm⁻³ and to 3SF (1)	Example of calculation: $28.80 \times 0.267 \div 1000 = 7.6896 \times 10^{-3} / 0.0076896 \text{ (mol)}$ $7.6896 \times 10^{-3} \div 3 = 2.5632 \times 10^{-3} / 0.0025632 \text{ (mol)}$ $2.5632 \times 10^{-3} \div 25 \times 1000 = 0.10253 \text{ (mol dm}^{-3}\text{)}$ $0.10253 \times 250 \div 25 = 1.0253 \text{ (mol dm}^{-3}\text{)}$ $1.0253 \times 192 = (196.86)$ $= 197 \text{ (g dm}^{-3}\text{)}$ Correct answer with no working scores 5. Ignore SF except 1 SF until final answer to 3 SF TE throughout and from (b)(i) Ignore units Steps can be done in any order	(5)

Question Number	Answer	Additional Guidance	Mark
3(b)(iii)	An answer that makes reference to the following points: there is no difference in effectiveness /effectiveness is (slightly) less (1)	Allow answers based on TE from (ii). Answer must be comparative for M1 if the calculated value is much less/much more than the correct value	(2)

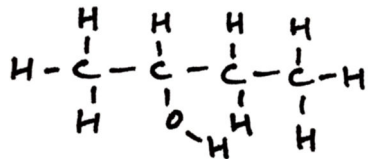
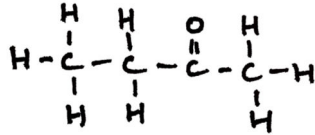
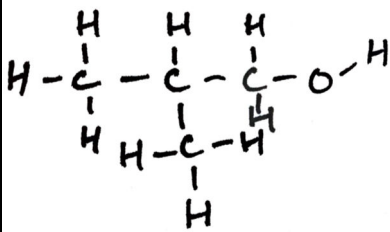
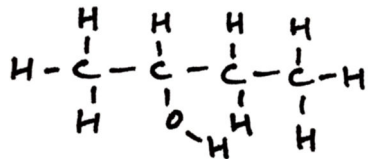
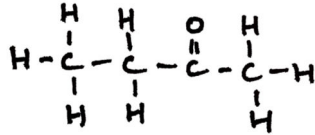
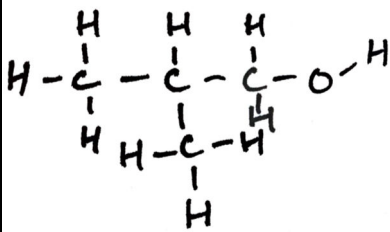
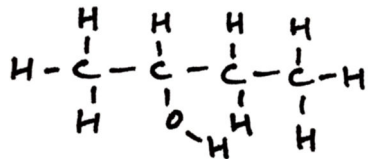
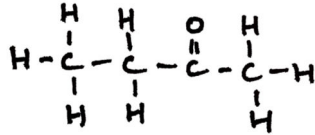
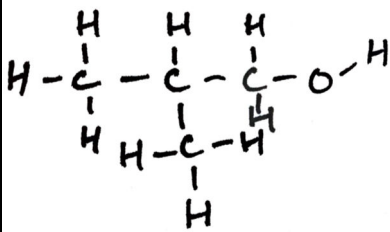
	the measured concentration of the descaler was almost the same as / slightly less / only ~1.5% less than the stated concentration	(1)	If candidates have failed to $\times 10$ then the measured concentration of the descaler is much less (so it would be much less effective) If candidates value is very high then concentration of descaler is much greater so might damage the appliance or be unsafe to use in the home.	
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(Total for Question 3 = 18 marks)

Question Number	Answer	Additional Guidance	Mark
4(a)	An explanation that makes reference to the following points: (in a dry test tube, add a small portion of) phosphorus(V) chloride / PCl_5 - steamy/misty fumes (would be seen)	M2 dependent on M1 or near miss Accept phosphorus pentachloride Ignore heating Ignore phosphorus chloride in M1 as a near miss Accept sodium Accept carboxylic acid (and H^+) and warm Do not award $\text{PCl}_5(\text{aq})$ Do not award acidified potassium dichromate(VI) Allow white fumes Do not award smoke unless ammonia used to test fumes If sodium used accept effervescence / bubbles If carboxylic acid used accept fruity smell	(2)

Question Number	Answer	Additional Guidance	Mark
4(b)(i)	An answer that makes reference to the following points: - orange (to) green	Allow blue	(2)

Question Number	Answer	Additional Guidance	Mark
4(b)(ii)	An answer that makes reference to the following point: (2-)methylpropan-2-ol / $(\text{CH}_3)_3\text{COH}$	Accept skeletal / displayed formula Allow methy- / methy- / methylpropane-2-ol	(1)

Question Number	Answer	Additional Guidance	Mark				
4(b)(iii)	An answer that makes reference to the following points: • formula of butan-2-ol • formula of butanone • formula of (2-)methylpropan-1-ol	Accept displayed / skeletal / structural formulae <table><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table> Penalise use of names once only If L is incorrectly shown as butan-1-ol then allow 1 mark for butanal or butanoic acid as the oxidation product.					(3)
							
							

Question Number	Answer	Additional Guidance	Mark
4(b)(iv)	An answer that makes reference to the following points: • add Fehling's or Benedict's solution (and warm) (1) • no reaction with (oxidation product) of L (1) • (brick) red ppt with (oxidation product) of M (1) OR • add Tollen's reagent/ammoniacal silver nitrate solution (and warm) (1) • no reaction with (oxidation product) of L (1) • silver mirror / black ppt with (oxidation product) of M (1) OR • add acidified sodium/potassium dichromate solution (1) • no reaction with (oxidation product) of L (1) • (orange to) green (solution) with (oxidation product) of M (1) OR • add iodine solution and aqueous sodium hydroxide (1) • (pale) yellow precipitate with (oxidation product) of L (1) • no reaction with (oxidation product) of M (1)	M2 and M3 are dependent on M1 or near miss No other tests than these four should be accepted. If L oxidation product is an aldehyde or missing, M2 is not awarded Allow orange If L oxidation product is an aldehyde or missing, M2 is not awarded If L oxidation product is an aldehyde or missing, M2 is not awarded Allow antiseptic smell If L oxidation product is an aldehyde or carboxylic acid or missing, M2 is not awarded	(3)

(Total for Question 4 = 11 marks)

(Total for Paper = 50 marks)